ALBERT-LUDWIGS-UNIVERSITÄT FREIBURG

ON THE EFFICIENCY OF EXCITONIC ENERGY TRANSPORT

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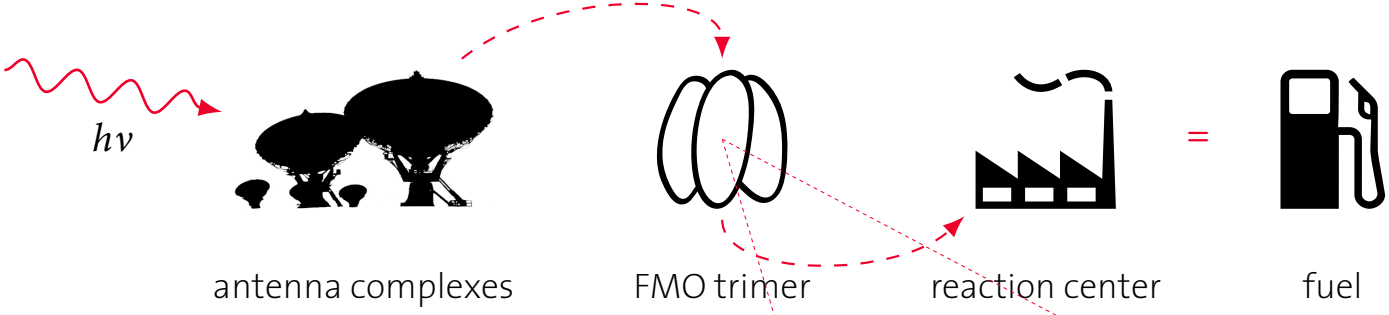
Introduction

Long-lived Quantum Coherence in Photosynthesis

experimental evidence[1,2] for quantum coherence in the Fenna-Matthews-Olson (FMO) protein of green sulfur bacteria

energy shuttle



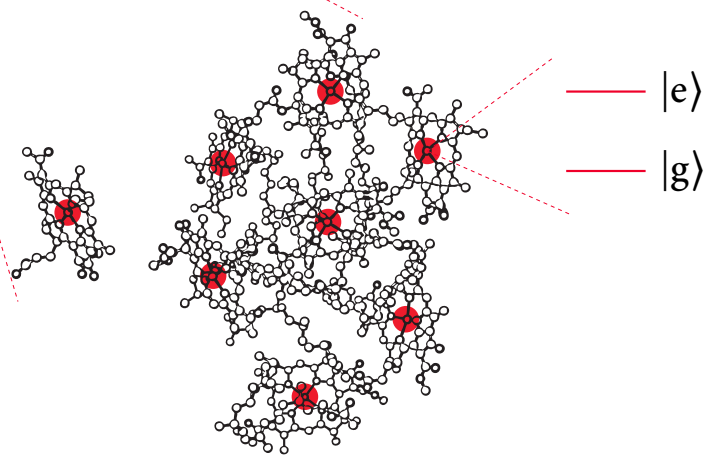
efficiency 95% of incoming energy reaches the reaction center

transport mechanism incentive to employ quantum coherence?

energy transport[3] mediated by 7 (or 8) molecular sites with discrete electronic energy levels

structural stiffness provided by surrounding protein scaffold

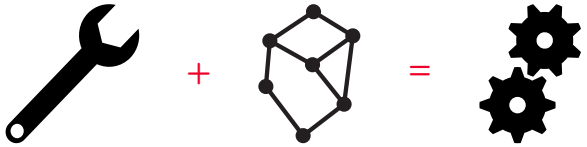
coherence preserved by nontrivial, noisy environment



REFERENCES [1]Engel et al., Nature, **446**, 782 (2007) ● [2]Panitchayangkoon et al., Proc. Natl. Acad. Sci. USA, **107**, 12766 (2010) ● [3]Adolphs et al., Biophys. J., **91**, 2778 (2006)

Candidate Mechanisms—The Role of Coherence

assistance of dephasing noise[4,5]	vs	enhancement by quantum interference[6,7]
noise suppresses detrimental destructive interference, *i.e.* quantum localization	**PREMISE**	interference sensitive to system parameters, fluctuations hallmark of coherence
optimize bath to foster energy transport through disordered system	**OBJECTIVE**	*optimize system* such that constructive interference enables efficient transport



REFERENCES [4]Rebentrost et al., J. Phys. Chem. B, **113**, 9942 (2009) • [5]Plenio et al., New J. Phys., **10**, 113019 (2008) • [6]T. S., F. de Melo, T. Wellens, F. Mintert, and A. Buchleitner, Phys. Rev. E, **83**, 021912 (2011) • [7]T. S., F. Mintert, T. Wellens, and A. Buchleitner, Semiconductors and Semimetals, **83**, 1 (2010)

Ensembles of Dipolar Networks

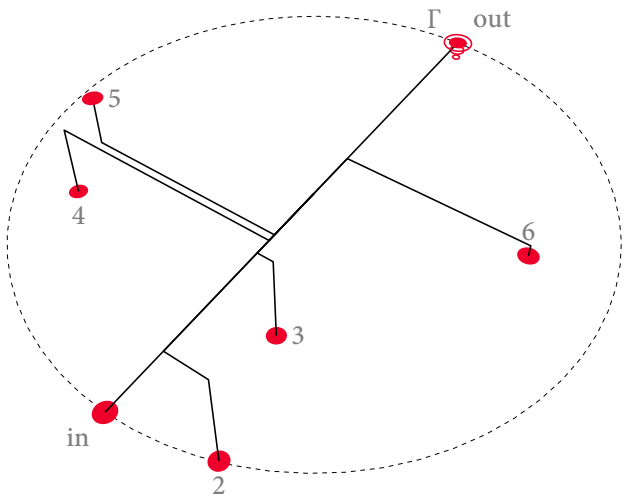
Spatial Disorder

single excitation transport from fixed node 'in' to fixed node 'out'

uniform distribution of intermediate nodes in d-ball, minimum distance $r_{\min} \leq r_{i,j} \leq r_{\text{in,out}}$

Hamiltonian $H = \sum_i E_i \, \sigma_{+,i}\sigma_{-,i} + \sum_{i \neq j} V_{i,j} \, \sigma_{+,i}\sigma_{-,j}$ with disordered, correlated $V_{i,j} = \mathfrak{A}\left(\boldsymbol{d}_i, \boldsymbol{d}_j, \vartheta_{i,j}, \varphi_{i,j}\right) r_{i,j}^{-3}$

reference time $T \equiv$ half a Rabi oscillation between $|\text{in}\rangle$ and $|\text{out}\rangle$



Ensembles of Dipolar Networks

Transport Efficiency

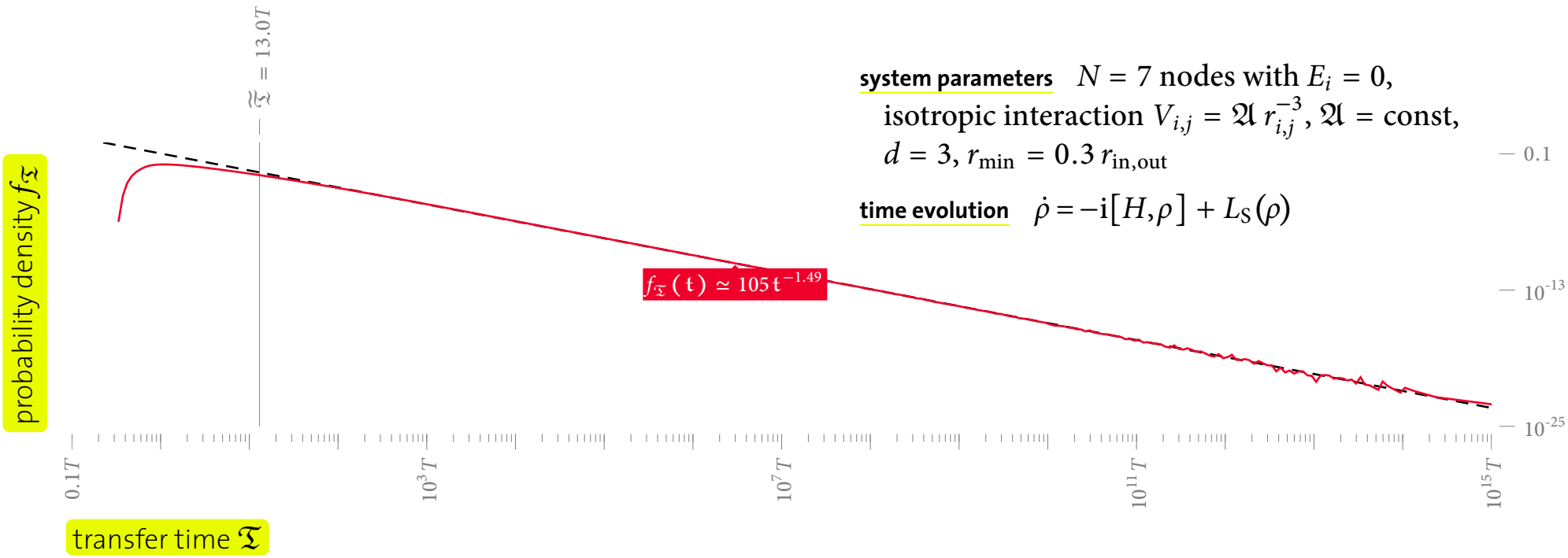
irreversible transfer to sink at 'out' with rate Γ described by $L_S(\rho) = \Gamma\left(\sigma_{-,\mathrm{out}}\,\rho\,\sigma_{+,\mathrm{out}} - \frac{1}{2}\left\{\sigma_{+,\mathrm{out}}\,\sigma_{-,\mathrm{out}}, \rho\right\}\right)$

sink drain $\mathfrak{p}(t) = \Gamma \int_0^t \langle \mathrm{out}|\rho(t')|\mathrm{out}\rangle\, \mathrm{d}t' \xrightarrow{t\to\infty} 1$

efficiency measure average transfer time $\mathfrak{T} = \int_0^\infty t\,\dot{\mathfrak{p}}(t)\,\mathrm{d}t$

heavy-tailed probability density $f_{\mathfrak{T}}(\mathfrak{t}) \propto \mathfrak{t}^{-\mu}$, $\mu > 1$, $\mathfrak{t} \to \infty$

typical value given by median $\widetilde{\mathfrak{T}}$ defined by $\int_0^{\widetilde{\mathfrak{T}}/T} f_{\mathfrak{T}}(\mathfrak{t})\,\mathrm{d}\mathfrak{t} = 1/2$



system parameters $N = 7$ nodes with $E_i = 0$, isotropic interaction $V_{i,j} = \mathfrak{A}\, r_{i,j}^{-3}$, $\mathfrak{A} = \mathrm{const}$, $d = 3$, $r_{\min} = 0.3\, r_{\mathrm{in,out}}$

time evolution $\dot{\rho} = -\mathrm{i}[H,\rho] + L_S(\rho)$

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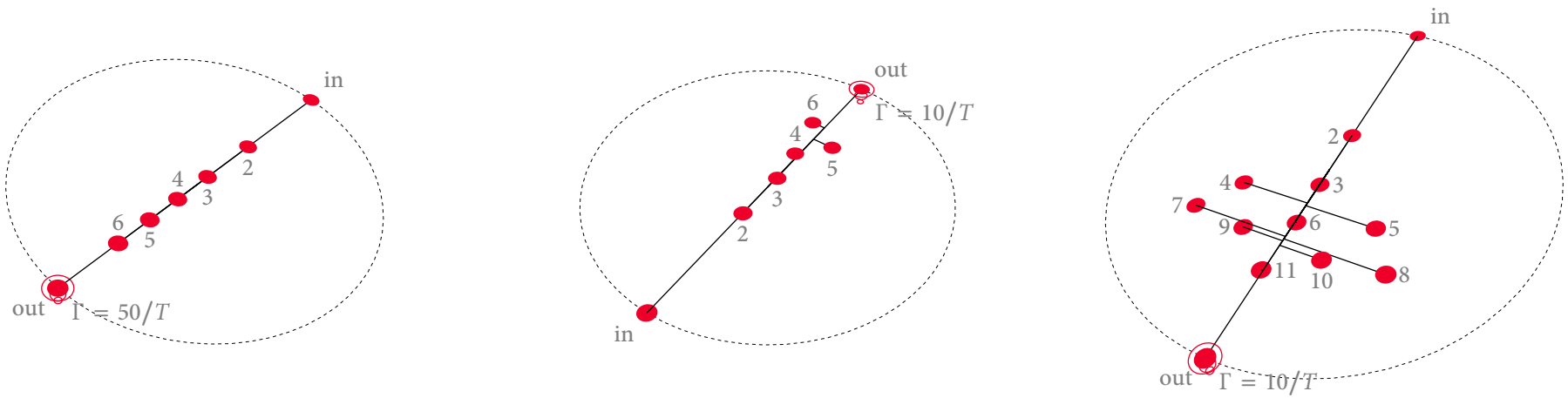
find optimal dephasing rates $\{\gamma_i\}$ for $L_{\mathrm{B}}(\rho) = 4\sum_i \gamma_i \left(\sigma_{+,i}\sigma_{-,i}\rho\,\sigma_{+,i}\sigma_{-,i} - \frac{1}{2}\{\sigma_{+,i}\sigma_{-,i},\rho\}\right)$	**CHALLENGE**	find optimal spatial configuration with $\mathfrak{T}_{\mathrm{opt}} \ll \widetilde{\mathfrak{T}}$



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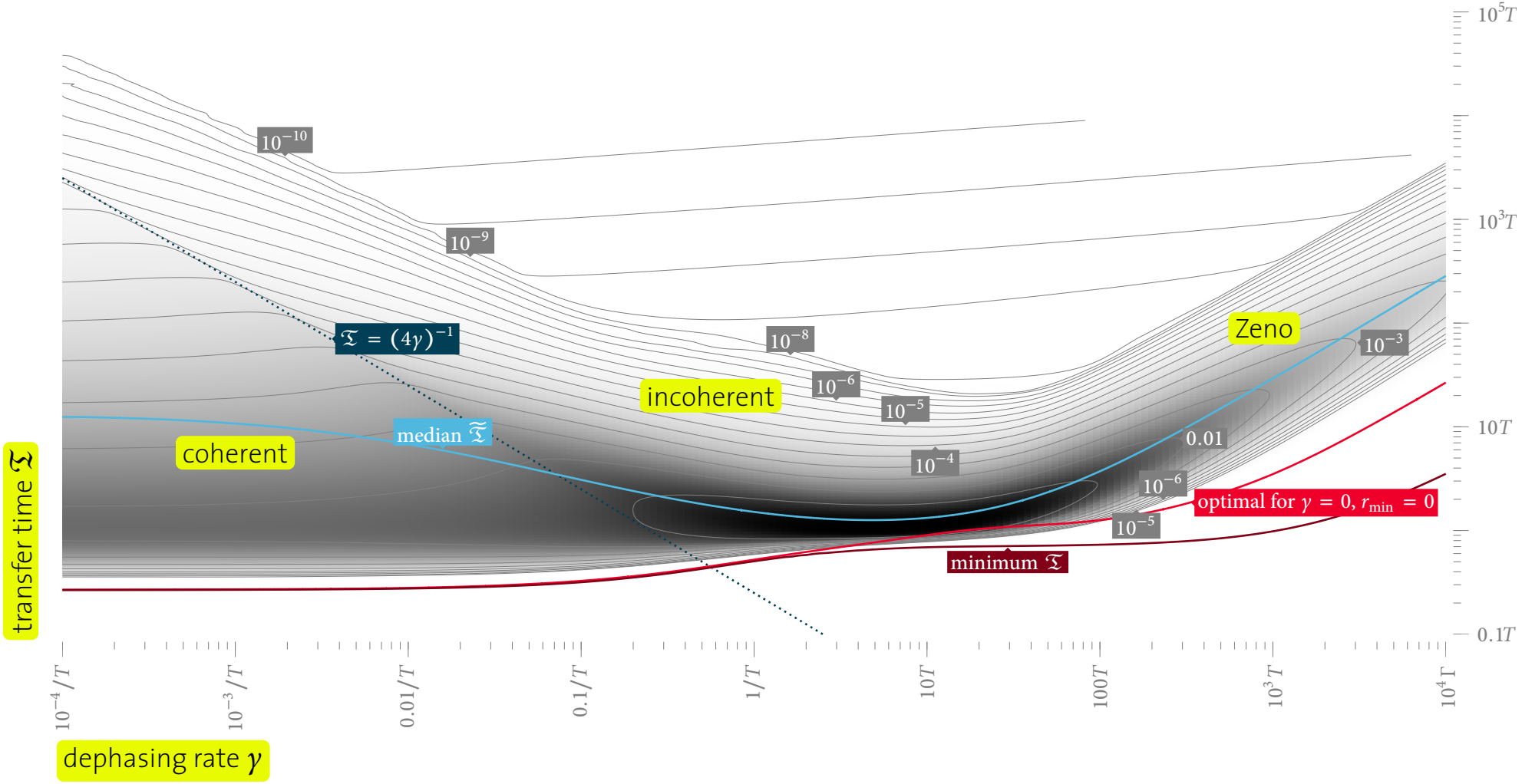
Topography of Transport And Dephasing

system parameters $N = 7$ nodes with $E_i = 0$, isotropic interaction $V_{i,j} = \mathfrak{A}\, r_{i,j}^{-3}$, $\mathfrak{A} = \text{const}$, $d = 3$, $r_{\min} = 0.3\, r_{\text{in,out}}$

time evolution $\dot{\rho} = -\mathrm{i}[H,\rho] + L_{\mathrm{S}}(\rho) + L_{\mathrm{B}}(\rho)$

timescales reference $T = \pi/(2|V_{\text{in,out}}|)$, sink $\Gamma^{-1} = T/10$, dephasing $(4\gamma)^{-1}$



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most configurations per se inefficient, can be assisted by noise at expense of coherence	**DIAGNOSIS**	optimal coherent transport of rare configurations *most efficient*[8,9]

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“THAT’S ALL FOLKS!”